Alberto Colombo

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Education

2017 - (ongoing)

M.Sc. - Politecnico di Milano, IT - Mathematical Engineering
Avg grade: 28.94/30

2015 - 2017

Bachelor - Politecnico di Milano, IT - Mathematical Engineering
110/110 cum laude

2009 - 2014

High school - ISISS L.Geymonat - Tradate, IT
100/100

Publications

2021

Positioning for high performance vehicles, *A. Colombo, U. Spagnolini*
Submitted to: *IEEE International Conference on Acoustics, Speech, and Signal Processing* (ICASSP) 2021

Employment

Aug. 2019 - (ongoing)

Math Tutor - English Business Services, Milano, IT
Part-time trainer for the quantitative reasoning section of graduate admission tests (GMAT, GRE, SAT).

Nov. 2018 - May 2019

Data Science Intern Campari S.p.A., Sesto S.Giovanni, Milano, IT
Intern in the new-born advanced analytics division.

July 2014

Flight Simulator Intern - Leonardo Helicopters, Sesto Calende, IT
Short internship in the flight simulator software team.

Projects

2020

M.Sc. thesis - Supervisor: U. Spagnolini
Title: Multiple sensors data fusion for positioning of high performance vehicles
Best Presentation Award at the 2019 Huawei Italy University Challenge (2° place)
Bayesian data fusion, localization, Kalman filter, Inertial measurement calibration.

2020

Advanced programming for scientific computing
Supervisors: L. M. Sangalli, L. Formaggia
Extension of the fdaPDE package.R software, C++, PDE, finite element methods, generalized linear models.

2019

Bayesian Statistics - Supervisors: A. Cadonna, A. Guglielmi
Spatio-temporal data, MCMC, non-parametric ANOVA, sparse prior.

2018

Applied Statistics - Supervisors: A. Menafoglio, P. Secchi
Compositional data, clustering, random forest, public speaking.
Statistical analysis of Mappa dei rischi dei comuni italiani.

2017

Bachelor thesis - Supervisor: E. Lanzarone
"Analysis of the predictive capabilities of electric current and pressure on the output of electrospinning process"
Collaboration with IMATI division of CNR on project nano-TWICE.

Skills

Coding

R:	*Expert - dplyr, shiny, ggplot2, stan.*
C++	*Highly proficient - eigen, expression templates, design pattern, MPI.*
MATLAB:	*Higly proficient - differential equations, signal processing, simulink.*
Python:	*Proficient - numpy, pandas, spark, matplotlib, sklearn.*
C, Pascal:	*Basic experience.*

Other softwares

Git, LaTeX, SQL, Linux(Ubuntu), Windows, Office.

Languages

Italian:	*Native speaker.*
English:	*Strong reading, writing and speaking. (TOEIC 2017: 845/990)*

Soft skills

Teaching, training, teamworking, personal communication, public performances.

Hobbies

Theatre, board-games, cooking, analog photography, hiking, biking.